

Digital Forensic Investigation using Autopsy

Tool: Autopsy

Theory

Autopsy is a free, open-source digital forensics platform used to analyse hard drives and disk images. It provides a graphical interface to browse file systems, recover deleted files, examine browser history and cookies, search keywords, and generate investigation reports. It is built on The Sleuth Kit (TSK) libraries and is widely used by law enforcement and security investigators.

Steps

1. Download Autopsy from <https://www.autopsy.com/download> and install it on Windows. Run the installer as Administrator and follow the setup wizard.
 2. Launch Autopsy. Click New Case on the welcome screen. Enter a case name such as EH_Lab_Case_01, set the base directory to the Desktop, and click Next. Enter a case number and examiner name, then click Finish.
 3. The Add Data Source wizard opens. Select Disk Image or VM File and click Next. Browse to your disk image file (.img or .E01) and select it. Set the timezone and click Next.
 4. On the ingest modules screen, enable the following modules: Hash Lookup, File Type Identification, Keyword Search, Web Artifacts, and Recent Activity. Click Next and then Finish. Autopsy begins analysing the image automatically.
 5. Wait for the ingest to complete. Progress is shown at the bottom of the screen. Once done, the left panel shows the Evidence Tree with all findings organised by category.
 6. To view browser history, expand Results > Extracted Content > Web History in the left panel. The right panel lists all visited URLs with timestamps.
 7. To view deleted files, expand the data source in the Evidence Tree and look for files marked with a red X icon. These are deleted files that Autopsy has recovered from unallocated space on the disk.
 8. To search for specific keywords, go to Tools > Keyword Search. Enter a word such as password or confidential and click Search. Autopsy lists all files containing that keyword.
 9. To generate a report, click Generate Report from the top toolbar. Select HTML Report, choose which results to include, and click Generate. The report is saved inside the case folder on the Desktop.
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Conclusion

Autopsy successfully analysed the disk image and recovered browser history, deleted files, and keyword-matched content. The generated report provides a structured summary of all digital evidence found, suitable for use in a formal investigation.